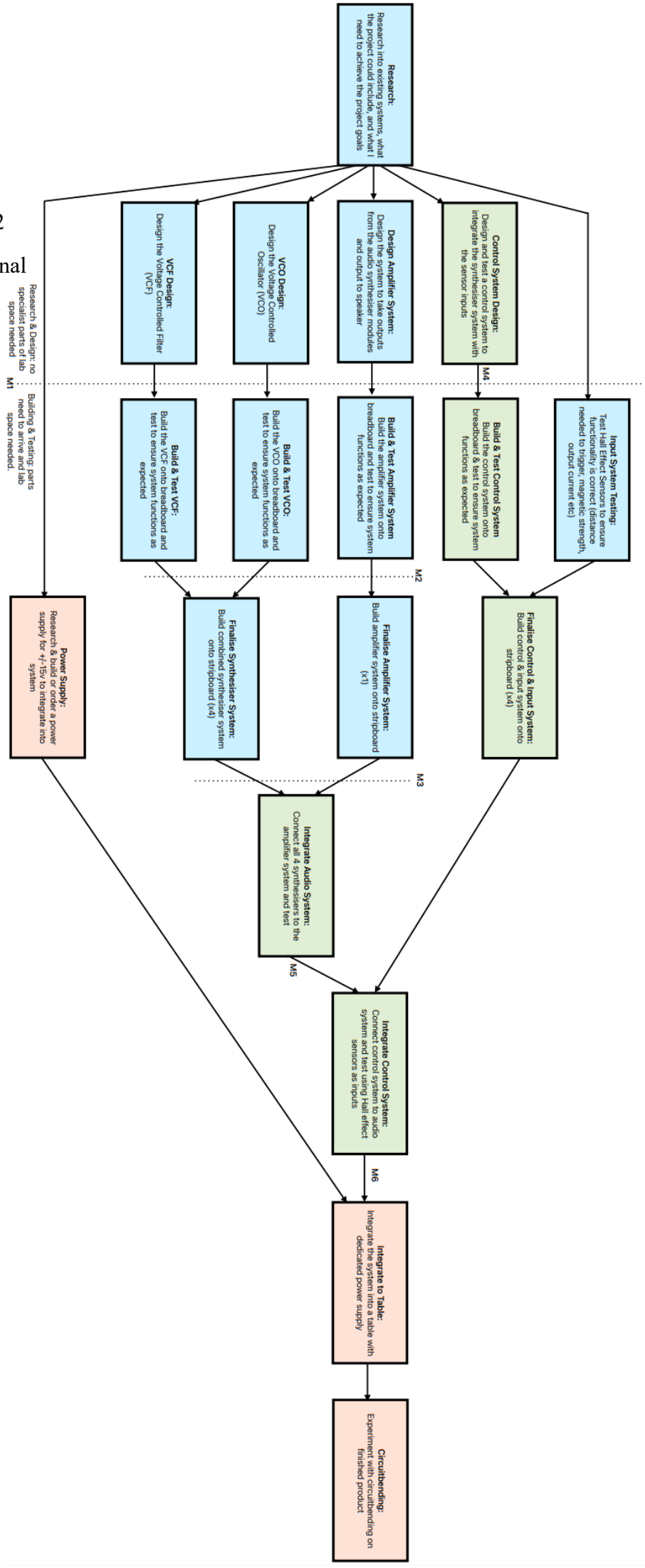
PERT Diagram:

Blue: Semester 1

Green: Semester 2

Orange: Aspirational



Milestones:

- **M1:** November 2025: Start on labwork
- **M2:** Early December 2025: Functioning amplifier and synthesiser system built onto breadboard
    - Amplifier system should be capable of losslessly summing outputs from all four synthesiser units with a user-controllable gain, which should have range of at least 1-10.
    - Amplifier should have suitable current output to allow for a low-impedance speaker to be connected – expected to be no less than 4 ohms across +/-15V.
    - VCO modules within synthesiser units should have controllable frequency and modulation inputs.
    - VCF modules within synthesiser units should be a low-pass filter with a voltage-controllable break frequency across the human hearing range with a maximum of 15V input.
    - Output performance of the synthesiser modules does not matter as they will be interfaced with the amplifier.
- **M3:** Late December 2025: Functioning amplifier and at least synthesiser system built onto stripboard (not yet integrated)
    - Functionality of systems should be unaffected from the versions built on breadboard.
- **M4:** Mid January 2025: System design for control system complete
    - Control system should be able to interface Hall-effect sensors with the synthesiser units. Exact performance is not yet determined due to the unknown inputs on the VCO.
- **M5:** Late January 2025: Integrate audio system (amplifier + synthesiser systems)
    - Performance should remain unaffected.
- **M6:** Late February 2025: Integrate control system to full system
    - Hall-effect sensors should be able to control every parameter available for the synthesiser modules to allow the system to be controlled from the backgammon board.

Stretch Goals:

- Power system: either purchase (short-time option) or design and build (long-time option) a power system (+/-15V) to allow the system to function without separate power supplies needed.
- Integrate the system into a table: mostly aesthetic work, allows the system to function as a fully standalone piece without specialist equipment or knowledge needed to operate it.
- Circuitbending: experiment with circuitbending within the control and synthesiser systems to further explore transient behaviours in the system.